Multimodal Real Estate Valuation: Integrating Satellite Imagery with Tabular Data

Soham Adak

January 5, 2026

Abstract

Traditional real estate valuation models rely on structured attributes such as square footage, construction quality, and geographic location. This project explores a multimodal regression framework that integrates tabular housing data with satellite imagery to capture environmental context. An automated pipeline was developed to fetch satellite images using geographic coordinates, and visual features were extracted using a pretrained ResNet-18 convolutional neural network. While a tabular Random Forest baseline achieved strong predictive performance ($R^2 \approx 0.88$), multimodal fusion experiments provided insights into feature redundancy and the complementary nature of visual neighborhood cues.

1 Introduction & Problem Statement

Real estate valuation is traditionally approached as a structured data problem. However, property prices are influenced by factors not fully captured in spreadsheets, such as neighborhood greenery, road density, and proximity to water bodies. Satellite imagery offers a scalable proxy for capturing such environmental characteristics.

This project investigates whether integrating satellite imagery with traditional tabular features can improve predictive accuracy. We design a multimodal learning pipeline combining structured housing attributes with CNN-extracted visual embeddings and rigorously compare its performance against a strong tabular baseline.

2 Dataset Description

2.1 Tabular Data

The dataset consists of historical housing transactions from King County, USA, comprising approximately 16,000 training samples. The target variable is `price`. Key predictors include:

- **sqft_living**: Interior living space.
- **grade**: Construction quality and architectural design.
- **sqft_living15**: Average living area of nearby properties (neighborhood wealth proxy).
- **latitude & longitude**: Spatial clustering and image acquisition.

2.2 Satellite Image Data

Satellite images were programmatically fetched using the Mapbox Static Images API. One image was collected per property using geographic coordinates at a fixed zoom level to balance property detail with neighborhood context such as greenery, road connectivity, and waterfront proximity.

3 Data Acquisition Pipeline

A dedicated script (`data_fetcher.py`) automated image retrieval. Engineering considerations included:

- Rate-limit handling to respect API quotas.
- One-to-one mapping between property IDs and images.
- Error handling for missing or duplicate entries.

4 Exploratory Data Analysis

4.1 Tabular EDA

Figure 1: Raw house price distribution showing heavy right skew.

Figure 1 shows a heavily right-skewed price distribution with high-value outliers, motivating a log transformation of the target variable.

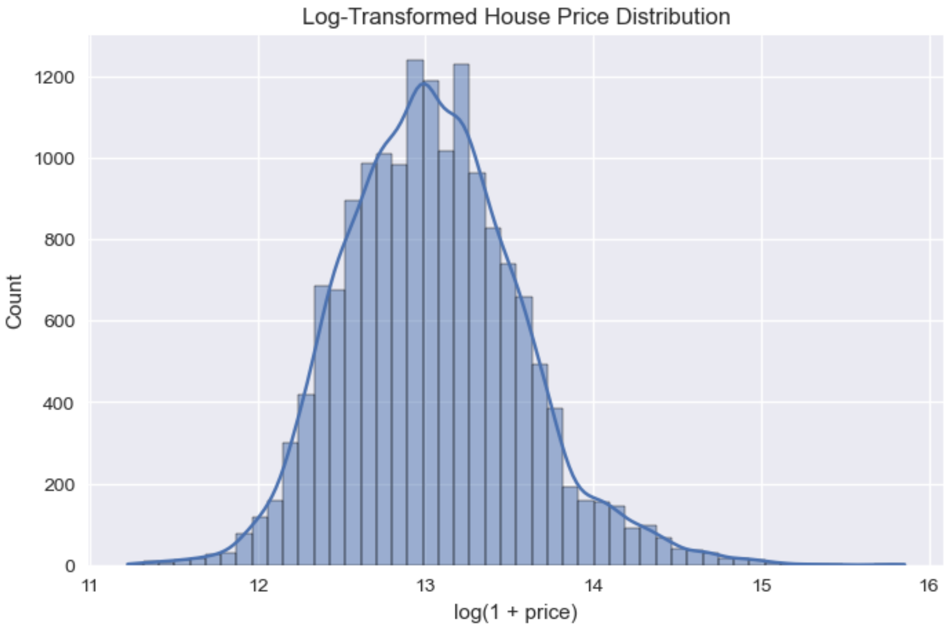

Figure 2: Log-transformed house price distribution.

The log-transformed prices (Figure 2) exhibit a near-normal distribution, improving regression stability.

Figure 3: Living area versus log-transformed house price.

Figure 3 highlights a strong positive correlation between living area and price.

4.2 Spatial Analysis

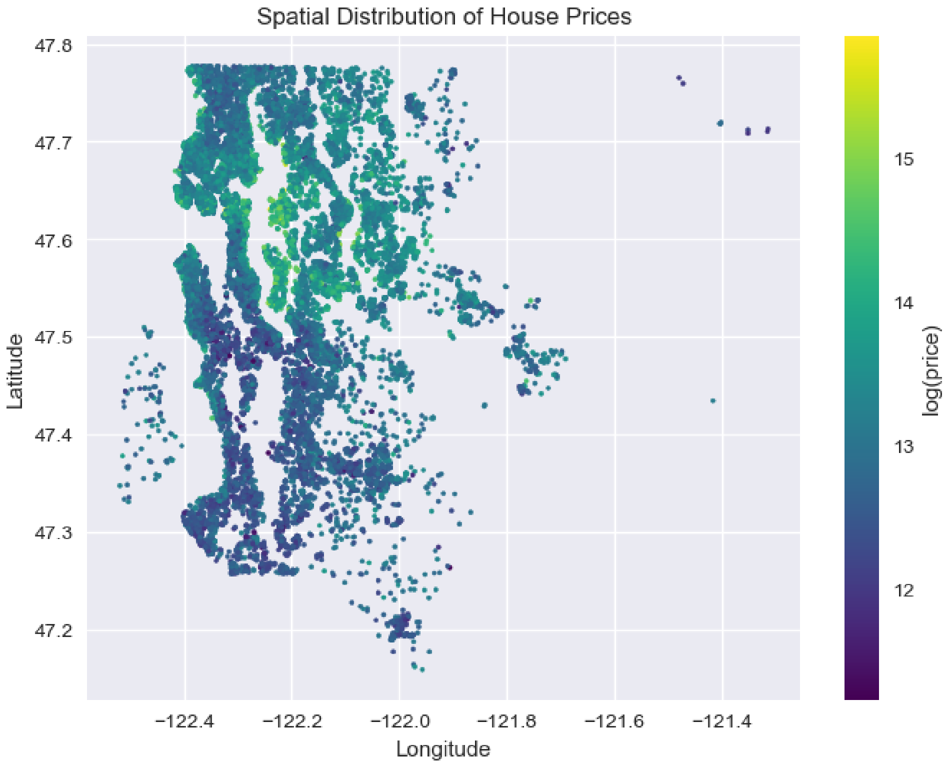

Figure 4: Spatial distribution of house prices across geographic coordinates.

Figure 4 reveals geographic clustering of high-value properties, particularly near waterfront regions and central urban zones.

4.3 Feature Correlations

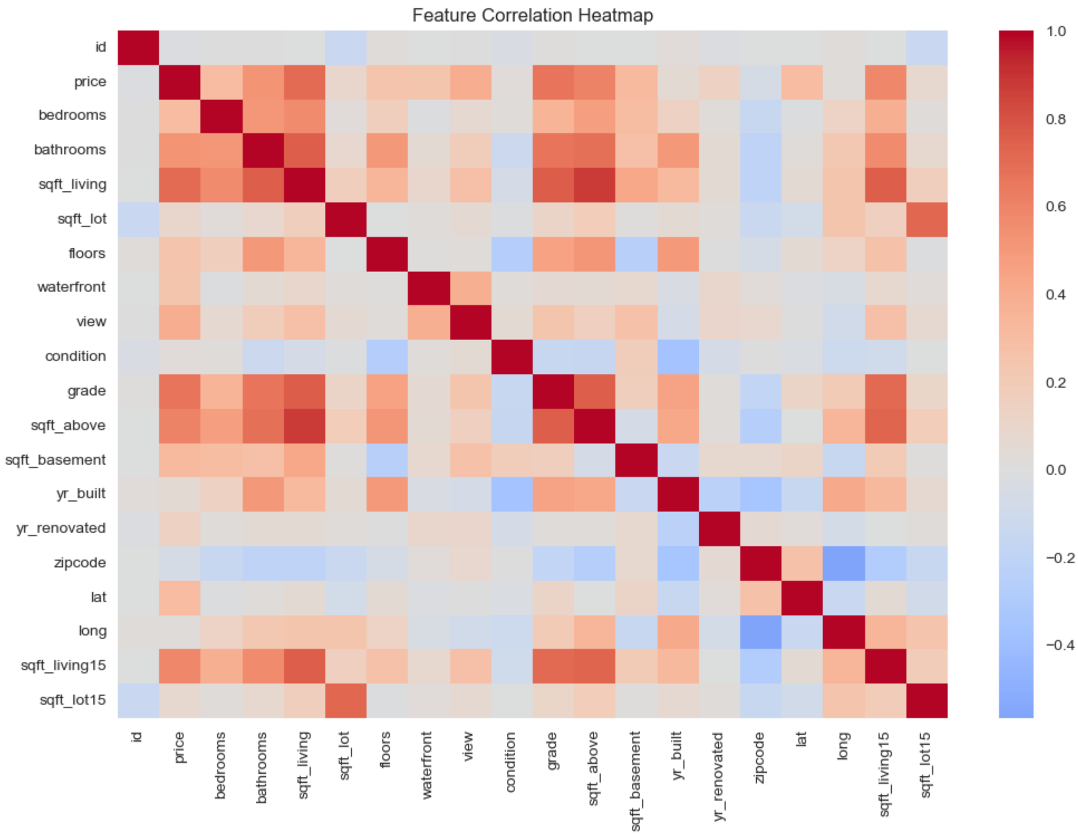

Figure 5: Feature correlation heatmap for tabular variables.

Figure 5 confirms that `sqft_living` and `grade` are among the strongest predictors of price.

5 Feature Engineering

5.1 Tabular Processing

Non-numeric columns were removed, duplicate property IDs were resolved, and the target variable was log-transformed to reduce skewness.

5.2 Image Feature Extraction

A pretrained ResNet-18 CNN was used as a fixed feature extractor. Images were resized to 224×224 and normalized using ImageNet statistics. The final classification layer was removed, yielding a 512-dimensional embedding for each image.

6 Modeling Approach

6.1 Baseline Model

A Random Forest Regressor trained on tabular features served as the baseline due to its robustness to non-linearities and strong performance on structured data.

6.2 Multimodal Fusion

Two fusion strategies were explored:

1. **Early Fusion with MLP**: Direct concatenation of tabular features and image embeddings, which resulted in overfitting due to high dimensionality.

2. **Tree-Based Fusion with PCA**: Image embeddings were reduced using PCA before fusion with tabular features and training a Random Forest.

7 Results & Evaluation

Model	**RMSE (log)**	R^2
Tabular Random Forest	0.1787	0.8843
Multimodal RF (PCA Fused)	0.1822	0.8727

Table 1: Performance comparison between baseline and multimodal models.

The tabular model achieved the best performance, indicating that structured features already capture most high-value signals. Satellite imagery provided complementary context but did not yield a significant predictive improvement.

8 Explainability & Visual Insights

Gradient-based visualization techniques such as Grad-CAM can highlight spatial regions influencing CNN activations. However, since the CNN was used as a frozen feature extractor rather than trained end-to-end for price prediction, Grad-CAM was not directly applied. Interpretability was instead achieved through qualitative visual analysis and comparative modeling.

9 Discussion, Limitations & Future Work

The dominance of tabular features limits the marginal contribution of visual data. Satellite resolution constraints and fixed zoom levels may also obscure fine-grained property details.

Future work includes fine-tuning CNNs on geospatial datasets, incorporating multi-scale imagery, and leveraging temporal satellite data.

10 Conclusion

This project implemented a complete multimodal valuation pipeline integrating satellite imagery with tabular housing data. While tabular features remained the strongest predictors, the study demonstrates a scalable framework for incorporating environmental context into real estate analytics.